COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 / Automating Server Management	**SECTION:**	CEP31S2
DATE PERFORMED:	7/16/2026	**DATE SUBMITTED:**	07/16/2026
NAME:	Villareal, Vanderohe Vernon	**INSTRUCTOR:**	Engr. Jimlord Quejado

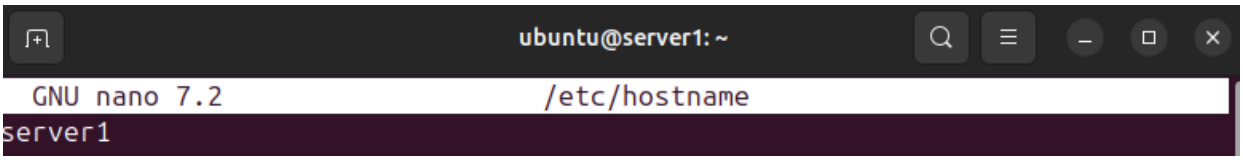
ACTIVITY NO. 1.1 **Creating Virtual Machines**

1. PROCEDURE OUTPUT

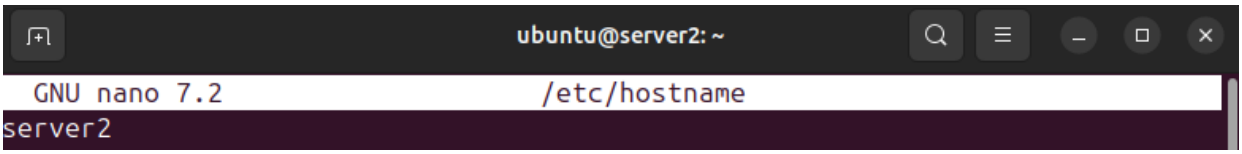
Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end

1. Change the hostname using the command sudo nano /etc/hostname

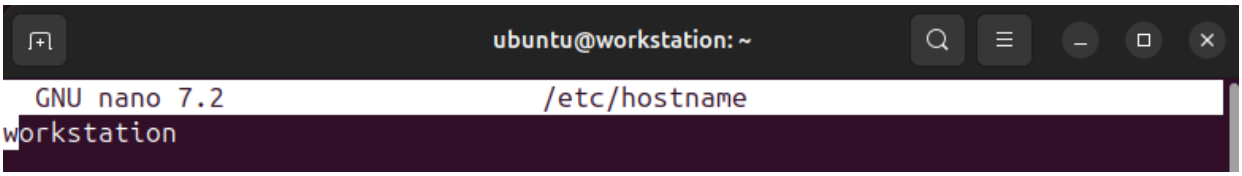
1.1 Use server1 for Server 1



1.2 Use server2 for Server 2

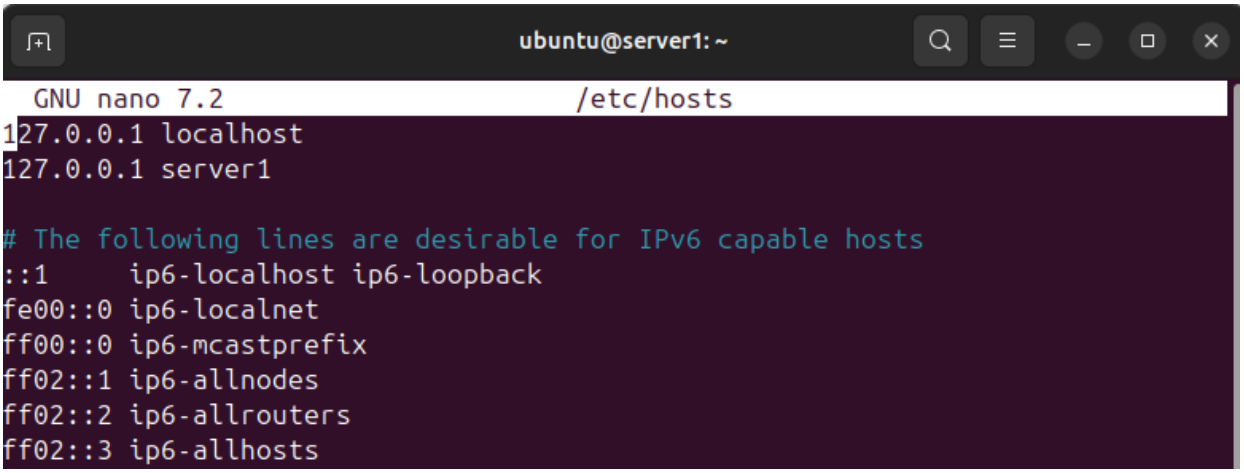


1.3 Use workstation for the Local Machine

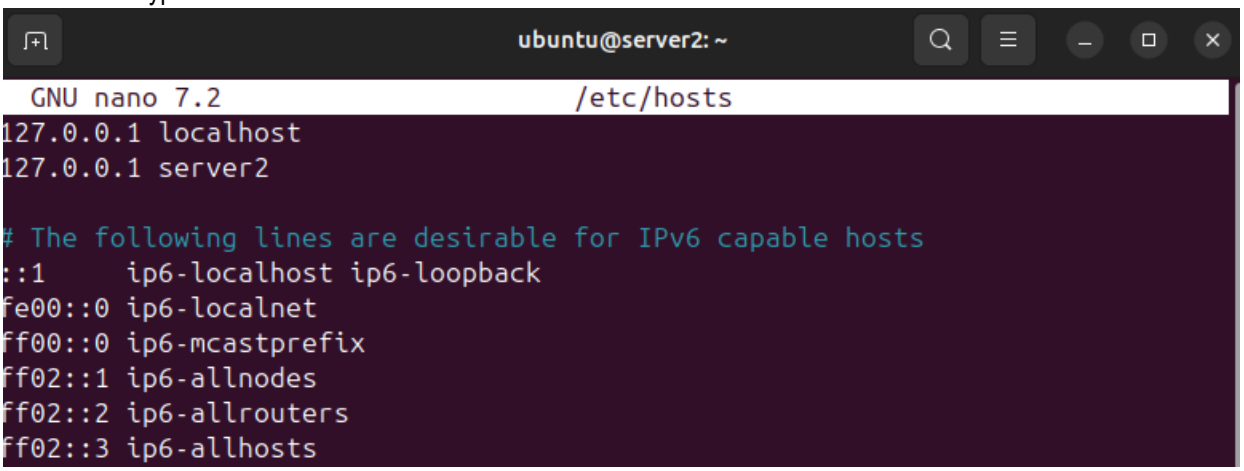


2. Edit the hosts using the command sudo nano /etc/hosts. Edit the second line.

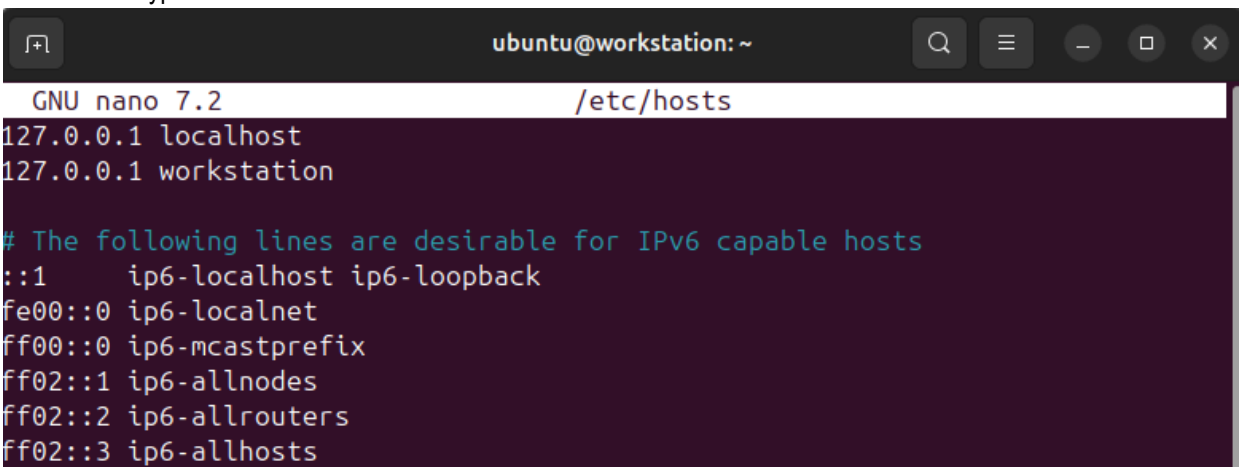
2.1 Type 127.0.0.1 server 1 for Server 1



2.2 Type 127.0.0.1 server 2 for Server 2



2.3 Type 127.0.0.1 workstation for the Local Machine



Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

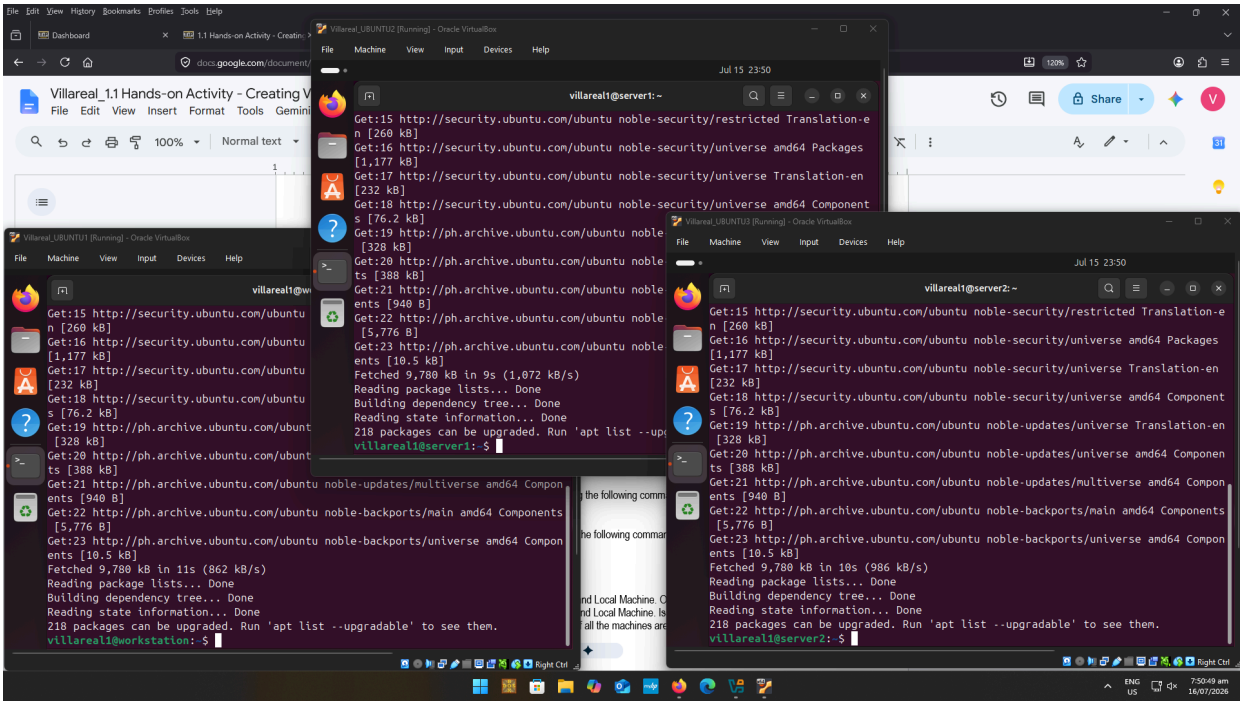
1. Upgrade the packages by issuing the command sudo apt update and sudo apt upgrade respectively.

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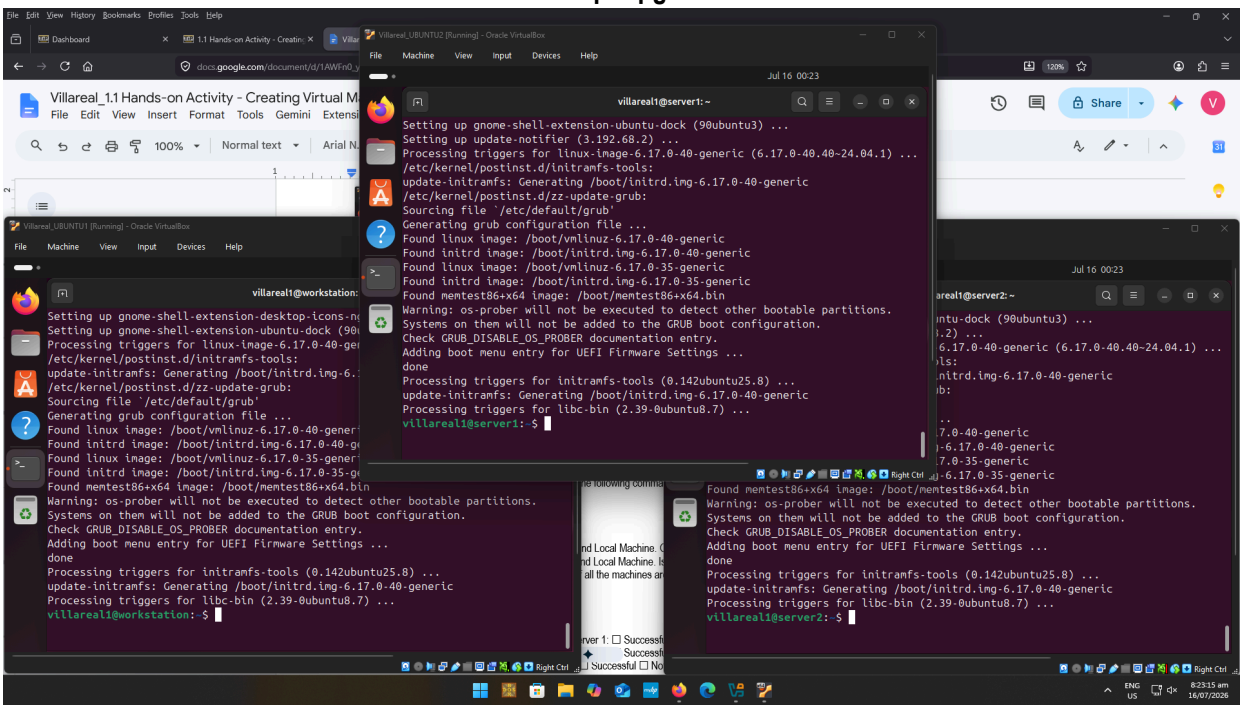
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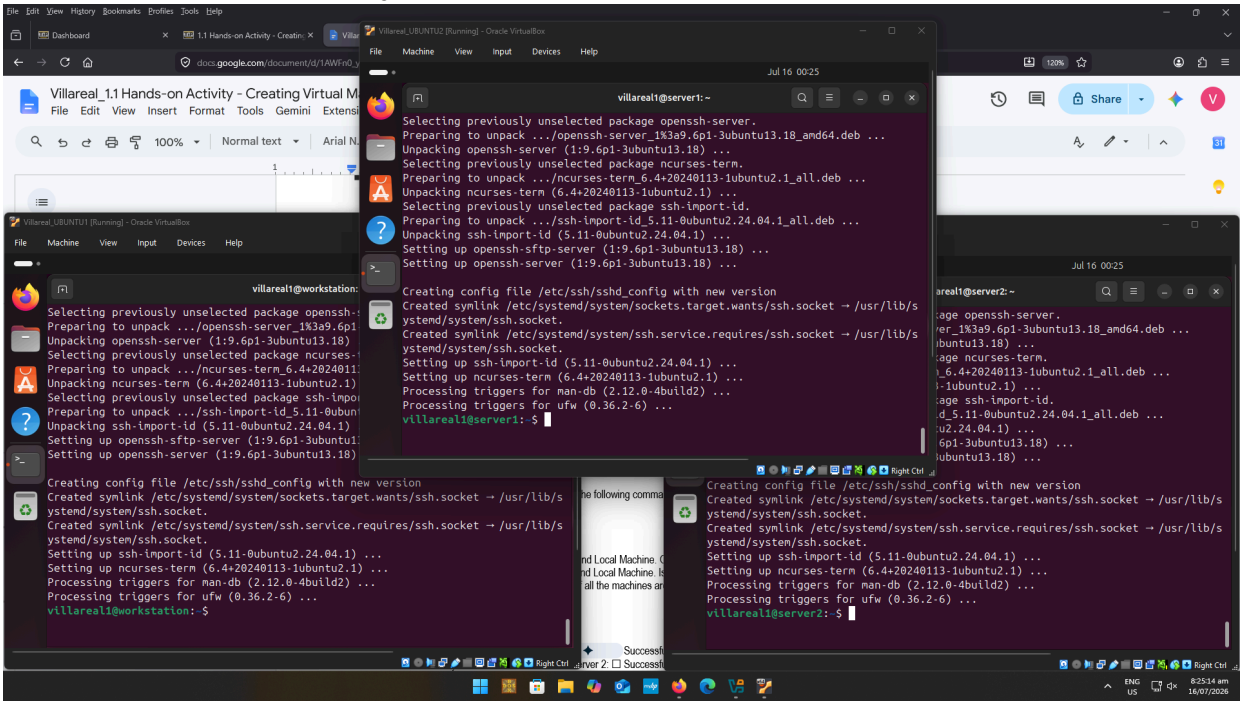
sudo apt update



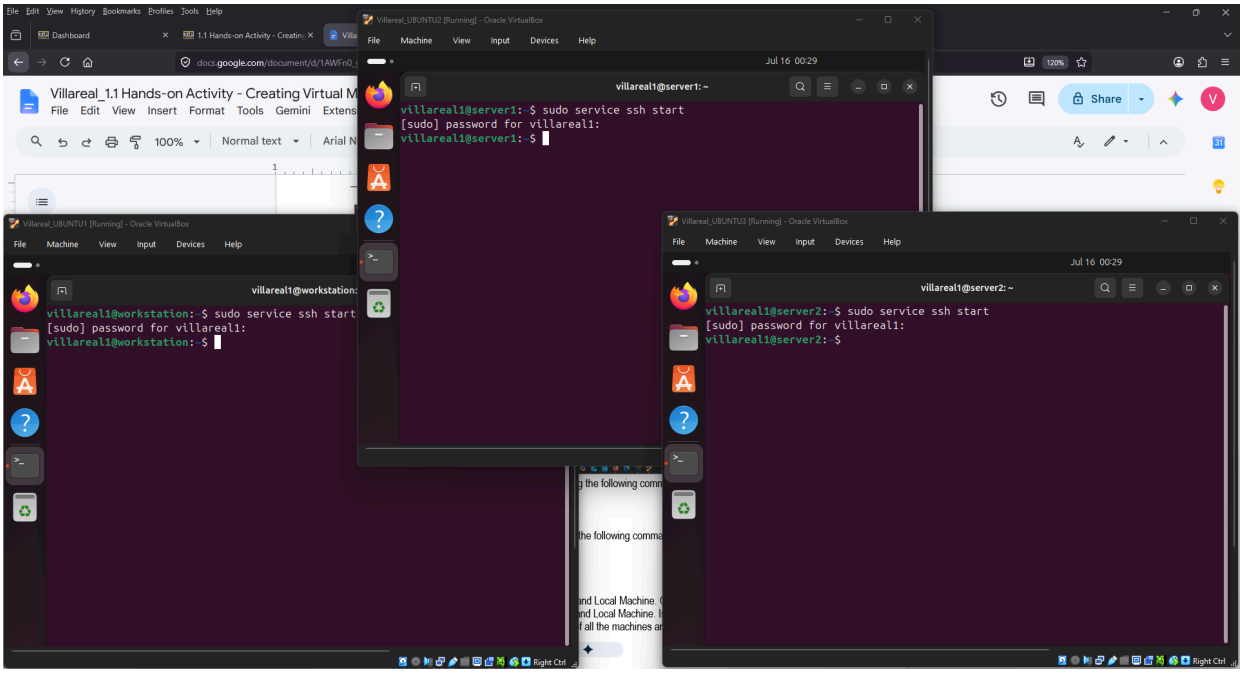
sudo apt upgrade



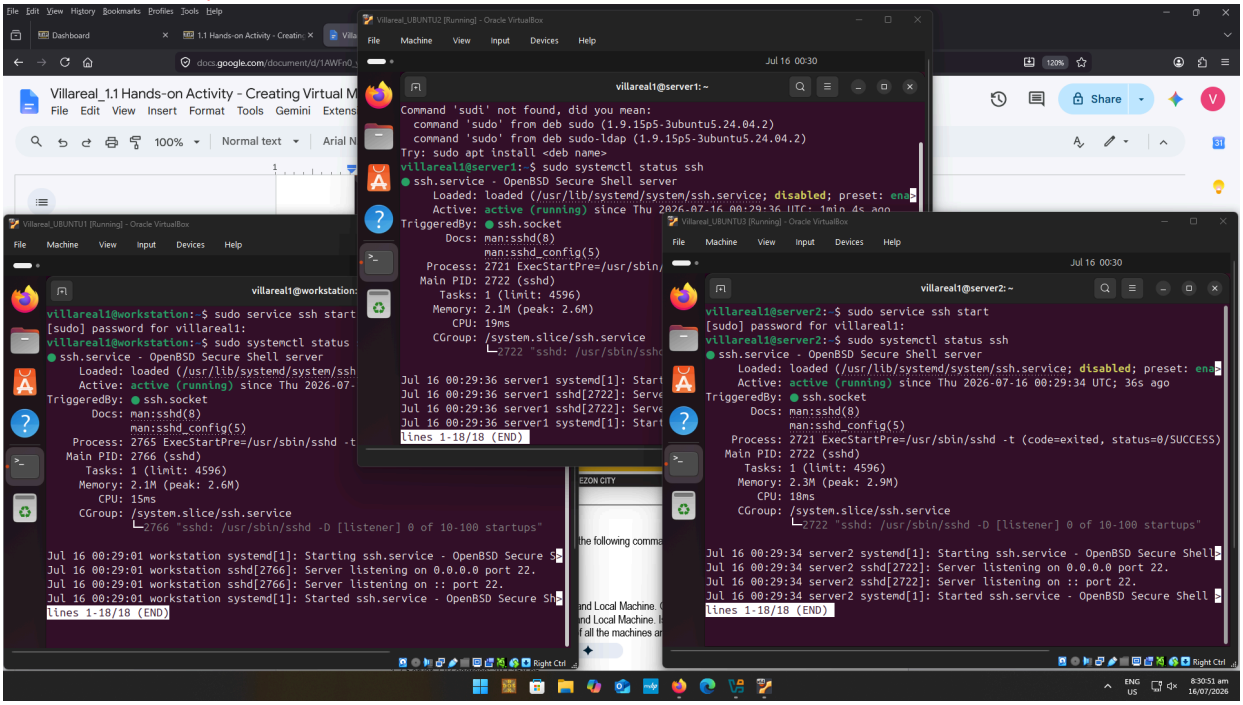
2. Install the SSH server using the command sudo apt install openssh-server.



3. Verify if the SSH service has started by issuing the following commands:
   3.1 sudo service ssh start

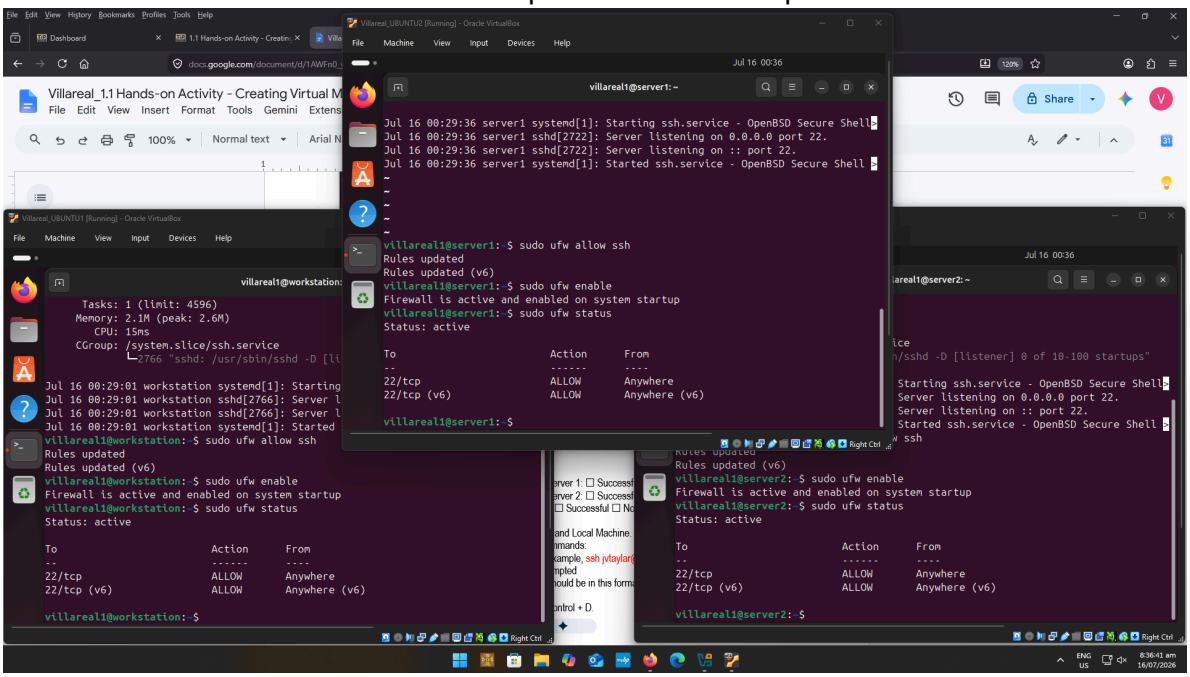


3.2 sudo systemctl status ssh



4. Configure the firewall to all port 22 by issuing the following commands:
   4.1 sudo ufw allow ssh
   4.2 sudo ufw enable
   4.3 sudo ufw status

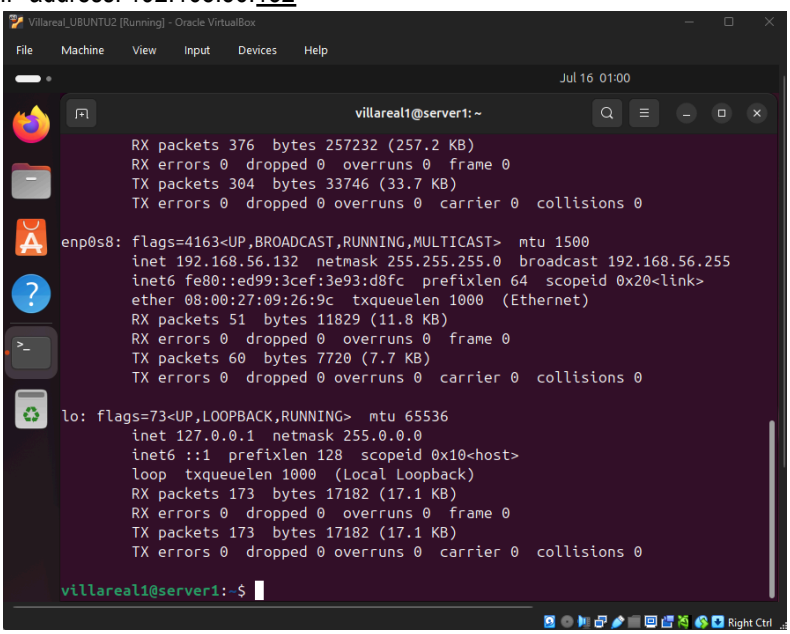
sudo ufw allow | ssh sudo ufw enable | sudo ufw status



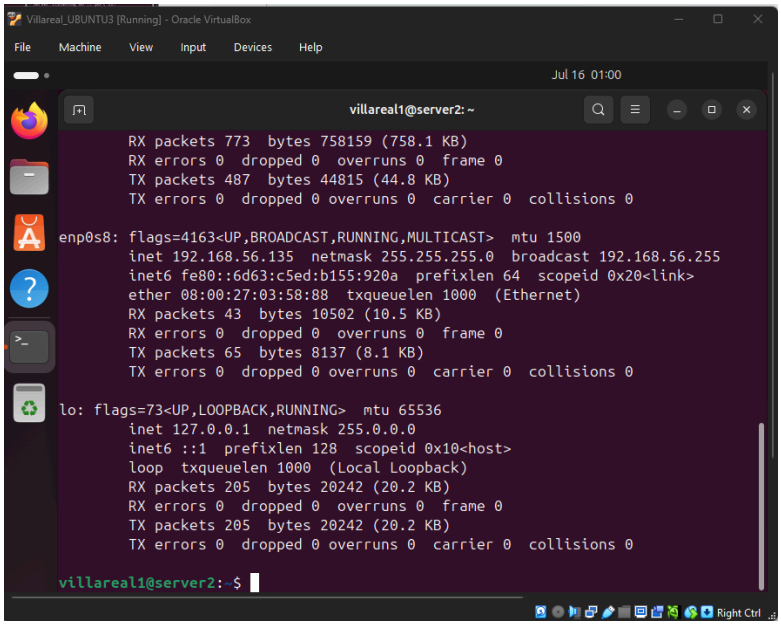
Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command ifconfig and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.
2.

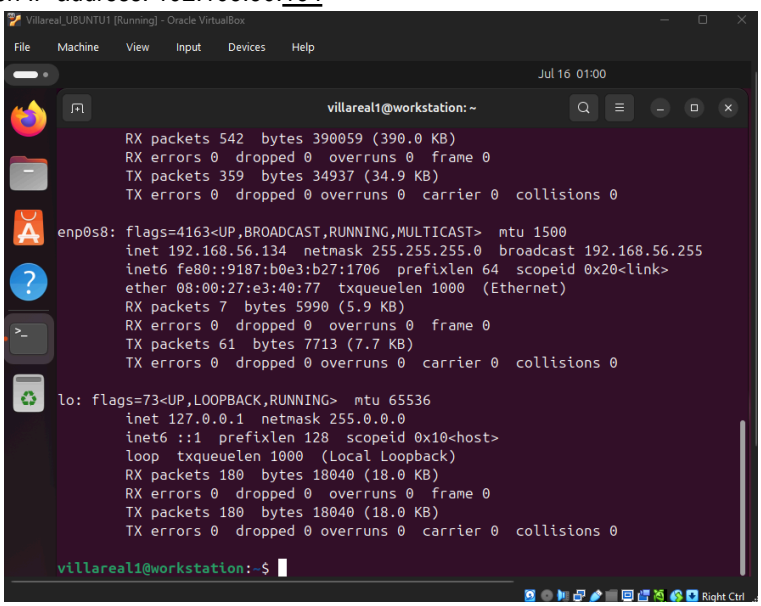
1.1 server1 IP address: 192.168.56.132



1.2 server2 IP address: 192.168.56.135



1.3 workstation IP address: 192.168.56.134



3. Make sure that they can ping each other.

2.1 Connectivity test for ***workstation*** to ***server1***:

- ☑ ~~Successful~~
- ☐ NotSuccessful

```
villareal1@workstation:~$ ping 192.168.56.132
PING 192.168.56.132 (192.168.56.132) 56(84) bytes of data.
64 bytes from 192.168.56.132: icmp_seq=1 ttl=64 time=1.23 ms
64 bytes from 192.168.56.132: icmp_seq=2 ttl=64 time=2.11 ms
64 bytes from 192.168.56.132: icmp_seq=3 ttl=64 time=1.10 ms
64 bytes from 192.168.56.132: icmp_seq=4 ttl=64 time=0.710 ms
^C
--- 192.168.56.132 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 0.710/1.287/2.106/0.510 ms
```

2.2 Connectivity test for ***workstation*** to ***server2***:

- ☑ ~~Successful~~
- ☐ NotSuccessful

```
villareal1@workstation:~$ ping 192.168.56.135
PING 192.168.56.135 (192.168.56.135) 56(84) bytes of data.
64 bytes from 192.168.56.135: icmp_seq=1 ttl=64 time=3.10 ms
64 bytes from 192.168.56.135: icmp_seq=2 ttl=64 time=1.21 ms
64 bytes from 192.168.56.135: icmp_seq=3 ttl=64 time=0.651 ms
64 bytes from 192.168.56.135: icmp_seq=4 ttl=64 time=1.13 ms
^C
--- 192.168.56.135 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3056ms
rtt min/avg/max/mdev = 0.651/1.524/3.104/0.936 ms
```

2.3 Connectivity test for ***server1*** to ***server2***:

- ☑ ~~Successful~~
- ☐ Not Successful

```
villareal1@server1:~$ ping 192.168.56.135
PING 192.168.56.135 (192.168.56.135) 56(84) bytes of data.
64 bytes from 192.168.56.135: icmp_seq=1 ttl=64 time=1.88 ms
64 bytes from 192.168.56.135: icmp_seq=2 ttl=64 time=1.60 ms
64 bytes from 192.168.56.135: icmp_seq=3 ttl=64 time=0.623 ms
64 bytes from 192.168.56.135: icmp_seq=4 ttl=64 time=0.456 ms
^C
--- 192.168.56.135 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3064ms
rtt min/avg/max/mdev = 0.456/1.139/1.876/0.610 ms
```

Task 4: Verify SSH connectivity on server1, server2, and workstation.

1. On the workstation, issue the following commands:

1.1 ssh username@ip_address_server1 for example, ssh jvtaylar@192.168.56.120

```
villareal1@workstation:~$ ssh villareal1@192.168.56.132
The authenticity of host '192.168.56.132 (192.168.56.132)' can't be established.
ED25519 key fingerprint is SHA256:MxkG9TvMHMeujQu1HsirYhHY5lyZ+T/NRzJPX5y6GzA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.56.132' (ED25519) to the list of known hosts
.
```

1.2 Enter the password for server1 when prompted

1.3 Verify that you are on server1. The user should be in this format user@server1.

```
villareal1@192.168.56.132's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

villareal1@server1:~$
```

For example, jvtaylar@server1

```
villareal1@server1:~$
```

2. Logout of server1 by issuing the command control + D.

```
villareal1@server1:~$
logout
Connection to 192.168.56.132 closed.
villareal1@workstation:~$
```

3. Do the same for server2.

```
villareal1@workstation:~$ ssh villareal1@192.168.56.135
villareal1@192.168.56.135's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
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Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

villareal1@server2:~$
```

```
villareal1@server2:~$
logout
Connection to 192.168.56.135 closed.
villareal1@workstation:~$
```

4. Edit the hosts of the workstation by issuing the command sudo nano /etc/hosts. Below all texts type the following:
   4.1 IP_address server1 (provide the ip address of server 1 followed by thehostname)

```
192.168.56.132 server1
```

   4.2 IP_address server2 (provide the ip address of server 2 followed by thehostname)

```
192.168.56.135 server2
```

   4.3 Save the file and exit.

```
  GNU nano 7.2                          /etc/hosts
127.0.0.1 localhost
127.0.1.1 Villareal1
192.168.56.132 server1
192.168.56.135 server2

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters









                        [ Wrote 11 lines ]
^G Help       ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit       ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

5. On the workstation, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do ssh jvtaylar@server1. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2

workstation - server1

```
villareal1@server1: ~                    villareal1@workstation: ~
villareal1@workstation:~$ ssh villareal1@server1
The authenticity of host 'server1 (192.168.56.132)' can't be established.
ED25519 key fingerprint is SHA256:MxkG9TvMHMeujQu1HsirYhHY5lyZ+T/NRzJPX5y6GzA.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
    ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server1' (ED25519) to the list of known hosts.
villareal1@server1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
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Expanded Security Maintenance for Applications is not enabled.

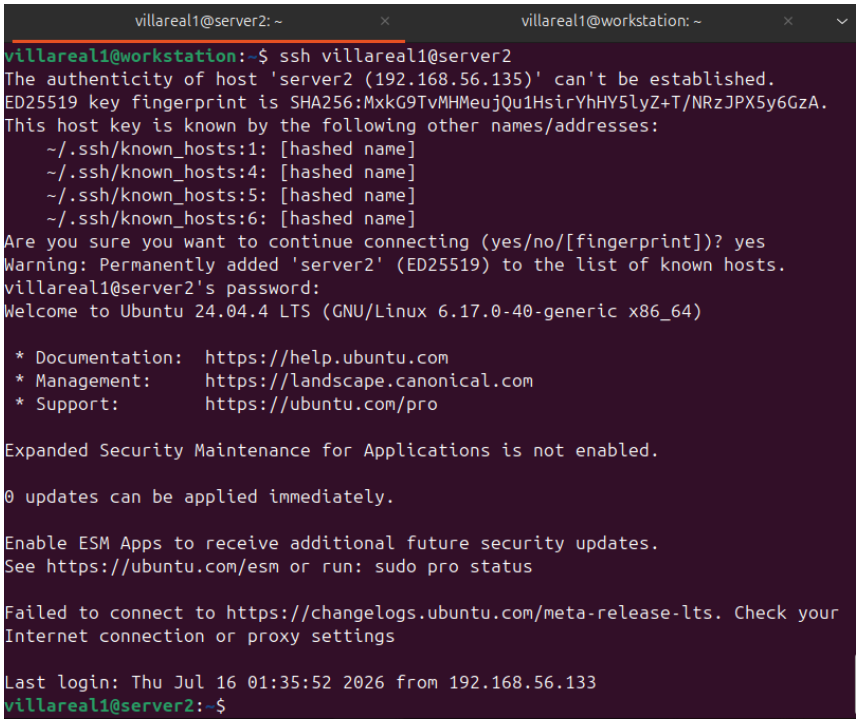
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings

Last login: Thu Jul 16 01:26:55 2026 from 192.168.56.133
villareal1@server1:~$
```

workstation - server2



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

1. How are we able to use the hostname instead of IP address in SSH commands?
   - **By adding the information of the other devices such as IP address and hostname into the" /etc/hosts" text file where it would be able to read it and ask for permission to use SSH commands on other devices using one VM.**
2. How secured is SSH?
   - **SSH is secure by default as it contains and uses encryptions that solely depend on how the user configures the authentications and accessibility of the machine. However, the security level of SSH drops when the attacker uses a brute force type breachment since if a password login is enabled the hacker can try all combinations that eventually lead to password breach.**

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of problems encountered, assist on solving the issue and idea generation.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Extent of Use: **Used to help which setting to change in order speed up the virtual machine's UBUNTU installation and generate ideas to answer text type questions.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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